

RPAT Curriculum

Section 2 - Fixed Wing Aircraft:

Lift and Drag Written Assignment

NOTE: *Although this assignment is open-book, you may NOT receive or provide help from/to other students. By submitting, you agree that you completed this assignment without help from others. If you help other students or receive help from others you will receive a zero on this assignment and will not be invited to participate in other ADDA courses.*

1. Imagine an aircraft that has taken off with a mass of 10,000 kg fully loaded, a coefficient of lift of 1.1, and a wing surface area of 40 m^2 . It is flying from Lilongwe International Airport in Malawi to the Kotoka International Airport in Ghana which is approximately 4325 km away. The air is calm and measures an air density of 1.22 kg/m^3 . (You may assume that aircraft mass is constant throughout the flight)
 - a. If the aircraft is flying in straight and level flight, what speed must the aircraft be flying at?
 - b. How long will the trip take if the aircraft is able to maintain that speed for the entire journey?
 - c. Now imagine that there is a direct headwind of 8 m/s that the aircraft experiences throughout its entire flight. Assuming that the aircraft is still flying straight and level, how long will the trip take now?

2. After some time, the same aircraft from the previous example has finished refuelling and is set to take-off on route to Bromma International Airport in Sweden which is an approximately 6175 km flight. All flight conditions are the same, but external tanks have been added to the aircraft for the longer flight; its mass is now 12,500 kg and it has a frontal area of 17.5 m^2 .

** Please use the frontal area for drag/thrust calculations and the wing area provided in Question 1 for lift/weight calculations*

- a. Again, at what speed must the aircraft be flying at during straight and level flight?
- b. If the aircraft's total coefficient of drag, C_D , is 0.15, how much thrust is being produced by the motor?
- c. Now knowing both the thrust and the nominal cruise speed of the aircraft, what is the amount of power being produced to achieve these results?
- d. Assuming that the external fuel tanks were responsible for an increase of 0.025 to the aircraft's total C_D , (so the original aircraft's $C_D = 0.125$, and its original frontal area was 16.5 m^2) how much more thrust is required to be produced with the addition of the fuel tanks?
- e. If for every 2500 Newtons of thrust produced, a kg of fuel was expended every minute, what is the total mass of fuel for both the flights from Malawi to Ghana, and Ghana to Sweden, at the take-off masses of both examples?